

A Triangular Actuating Device Stand that Dynamically Adjusts Mobile Screen's Position

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ABSTRACT

This demo presents a triangular actuating device stand that can automatically adjust the height and tilt angle of a mounted mobile device (e.g., smartphone) to adapt to the user's varying interaction needs (e.g., touch, browsing, and viewing). We employ a unique mechanism to deform the stand's triangular shape with two extendable reel actuators, which enables us to reposition the mobile screen mounted on the hypotenuse side. Each actuator is managed by the mobile device and controls the height and base of the stand's triangular shape, respectively. To demonstrate the potential of our new actuating device stand, we present two types of interaction scenarios: manual device repositioning based on the user's postures or gestures captured by the device's front camera and automatic device repositioning that adapt to the on-screen contents the user will interact with (i.e., touch-based menus, web browsers, illustrator, and video viewers).

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

mobile device stand, shape-changing interface, ergonomics

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1 INTRODUCTION

Mobile touchscreen devices such as smartphones and tablets are wide spread and commonly used. They are used in various places

due to their high mobility and intuitive operation; furthermore many people have started using these devices as their primary workstation. However, such mobile devices generally need to be held by hands in the air, forcing the user to maintain an uncomfortable posture. For example, if the device is held at a low position, the risk of text-neck syndrome might increase due to the tendency to fall into a slumped posture [3, 11]. Arm fatigue is another issue that may occur if the device is held at eye level. A simple solution regarding device holding is to use a desktop device stand that can keep the mobile device's position at an appropriate height and angle, which can later be flexibly adjusted to adapt to their on-going interaction styles. However, these devices are generally designed as two types: *view priority* positions the screen high and vertical for browsing-centered tasks, and *interaction priority* positions the screen low and flat for activities such as sketching or touch. Furthermore, most existing device stands' arm joints can only be adjusted manually. Therefore, such conventional device stands cannot effectively adapt a user's mobile workstation to their varying tasks, ranging from touching/drawing (e.g., on sketching software) to viewing (e.g., video viewer) experiences.

This demo presents a novel triangular actuating device stand that can automatically adjust the height and tilt angle of a mounted mobile device (e.g., smartphone) to adapt to the user's interaction needs (e.g., touch, browsing, and viewing). The core idea is to make a triangular stand configurable by flexible and extendable reel actuators inspired by prior shape-changing devices [5, 6, 8, 10]. This triangular configuration with two actuators allows us to flexibly and dynamically control the triangle's height and base sides, enabling repositioning of the mounted mobile screen on the hypotenuse side according to the desired interaction activities (i.e., touching or viewing). In this demo, we show the potential of our shape-changing device stand by demonstrating two types of interaction scenarios: manual device repositioning and automatic device repositioning.

2 RELATED WORK

Many attempts have been made to optimize device positions for more comfortable and appropriate workspaces. Examples include moving displays while detecting the user's posture [7] and robotic workspaces which dynamically change the position of furniture or devices [1, 4, 9, 12]. However, these works are designed for desktop environments with regular monitors and do not accommodate for a mobile workplace with touchscreen devices.

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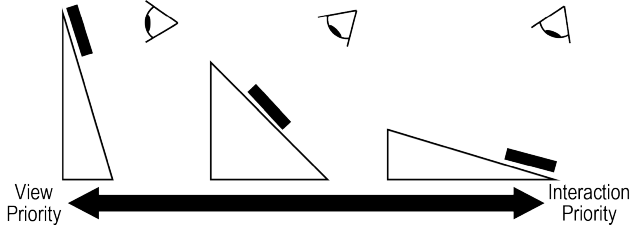


Figure 1: Different stands by shape-changing triangle

While there are many approaches to creating shape-changing physical structures, we focus on extendable reel actuators, inspired by several prior works designed for VR proxy [8], 3D display [10], and height-changing avatar [5]. We employ the extendable reel actuators with a different configuration to achieve a shape-changing desktop device stand for optimizing a workstation with a mobile device.

3 TRIANGULAR ACTUATING DEVICE STAND

3.1 Design

We aim to design a desktop device stand that can seamlessly transition between view priority and interaction priority to support various needs during mobile interactions. Figure 1 depicts side views of the triangular device stand and the mobile device can be fixed on the inclined side (as shown as a black plate). For view priority (Figure 1 left), the mobile device is positioned vertically at a higher level (e.g., eye level) to support continuous content viewing experience. In contrast, interaction priority (Figure 1 right) is generally helpful to accommodate the user’s touch interactions (e.g., sketch) on the screen without arm fatigue caused by maintaining prolong mid-air hand postures. To effectively support both modes, we propose a shape-changing triangular desktop device stand (Figure 1) whose shape can be automatically controlled by physically extending all three sides with reel actuators. Through a triangular shape-changing process, the mobile device’s position and angle changes from the high/vertical to low/flat modes. Compared to conventional arm-based stand, our mechanism would offer more scalable shape-changing possibilities with simpler and fewer actuators.

To adjust for view priority (Figure 1 left), we configure a right-angled triangular stand with a longer height and a short base side, where the device is fixed at a higher area of the inclined side. Based on prior ergonomics knowledge [2], we set the maximum height of the mounted device to 50 cm so that its screen height is approximately identical to the user’s eye level. To configure for interaction priority for sketching or touch activities, a flatter triangle (e.g., horizontally longer one) should be used, which can be created from the previous configured high right-angled one (Figure 1 left) by shrinking the height and extending the base. Figure 1 middle shows an isosceles triangular stand as their intermediate mode, which may be effective for browsing activities (e.g., web page browsing, book reading) involving both viewing and touch interactions.

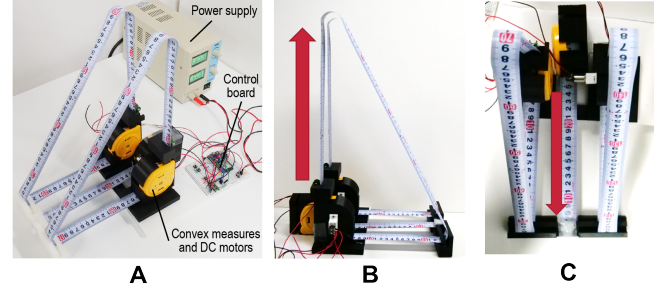


Figure 2: (A) Prototype overview (B) The two outer actuators extend the right triangle’s height and total length. (C) The inner actuator stretches the base side.

3.2 Implementation

We used a set of convex measure reels to implement our proof-of-concept prototype. Such reels have two unique properties: 1) extendable and maintains their shape while supporting modest loads in certain directions, and 2) the whole body is retractable with a coil spring. Similar to prior works [5, 6, 8, 10], we augment these properties by motorizing these measure reels.

Our prototype device (Figure 2 (A)) consists of three reel actuators. Each of the three convex measures is actuated by a DC motor (Pololu 1000:1 Micro Metal Gearmotor HPCB 6V), and each reel’s length is controlled by the amount of rotation of the DC motor. The motors are controlled by a motor driver (Pololu DRV8835 Dual Motor Driver Carrier), a microcontroller (ESP32-DevKitC-32E), and a DC 6V power supply (AD-8723D). The microcontroller is operated by commands from the mobile device through a wireless bluetooth connection. The outer two reels are placed vertically, which can form a basic right triangle shape and extend its height (Figure 2 (B)). Figure 3 (A) shows the stand in view priority, where the outer reels are extended (on the height). To create flatter triangular shapes, we introduced an additional inner reel arranged horizontally between the two outer reels, which allows us to change the length between the corner and the vertex of the base side of the triangular stand (Figure 2 (C)). Figure 3 (B) depicts interaction priority, where the device is flat and closer to the user (compared to view priority), which enables the user to more comfortably interact with the device for touch-based interactions. Figure 3 (C) is the initial and minimum mode of the stand. Our system’s various shapes can be reconfigured from this initial mode by simply controlling these three actuators.

4 DEMO APPLICATIONS

We present two demos, one of which is manual device repositioning where the user can manually adjust the position of the device (i.e., adjust the shape of the triangle) based on the view of the device’s front camera. The second one automatically adjusts the device position based on what content the user will interact with. Specifically, we focus on how frequently the user will touch the screen. Depending on the touch interaction frequency, a potentially suitable device position and angle (i.e., upright or flat) could be decided.

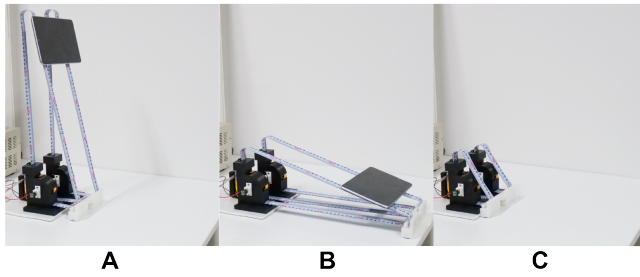


Figure 3: (A) view priority. (B) interaction priority. (C) Minimum mode.

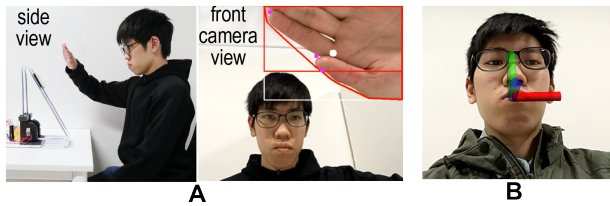


Figure 4: (A) The mounted mobile device's front camera detects the user's hand position (e.g., upper or lower), which can be used to manually raise or lower the device position. (B) The facial position or angle can also be a trigger to adjust the device's height.

4.1 Manual device repositioning

The front camera facing the sitting user might detect the user's face and upper body. This setup allows the user to manually adjust the device height and angle using hand or body gestures. Figure 4 (A) shows a user raising the mobile device with a hand-raising gesture. The user's posture and gaze direction captured from the front camera can also be used to reposition the device manually.

4.2 Automatic device repositioning

The system can automatically reposition the mounted device based on situational awareness technology. For simplicity, we focus on the content that the user is currently viewing on-screen which can be detected easily in the mobile device software; this information is critical to how the device should be positioned. If the user wants to sketch or engage in content exploration where many touch interactions on the surface are required, the device should be positioned flatter to make their hand posture more comfortable and their writing experience more effective. On the other hand, if the content is mainly for viewing, the device should be located at a higher position (e.g., slightly lower than the eye level). Figure 5 shows a scenario where the sitting user decided to watch a video, so the device screen is automatically adjusted to a higher position, allowing the user to maintain a back straight posture. As the automatic actuation might sometimes fail, the user can also use manual repositioning to optimize the device's position during or after the automatic device actuation.



Figure 5: Automatic device control that reposition the mobile device according to the screen content the user is viewing.

In conclusion, this demo will contribute a concept of a new actuating device stand that uses a triangular shape-changing mechanism, enabling dynamic reconfiguration of the mobile device's height and angle that best matches the on-going interaction activities. This allows us to always have an ergonomically comfortable mobile workspace for dynamically varying tasks between touching and viewing activities with mobile devices.

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